
Advanced Computer Architectures

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Name
Last Name
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Problem 1 (5 points)	
Problem 2 (5 points)	
Problem 3 (6 points)	
Question 1 (5 points)	
Question 2 (6 points)	
Question 3 (5 points)	
Total (32 points)	

Problem 1

In this problem, you will port code to a simple VLIW machine, and modify it to improve performance. Details about the VLIW machine:

- Three fully pipelined functional units (Integer ALU, Memory, and Floating Point)
- Integer ALU has a 1 cycle latency
- Memory Unit has a 3 cycle latency
- FPU has a 3 cycle latency and can complete one add or one multiply (but not both) per clock cycle
- No interlocks

C Code:

```
for(int i=0; i<N; i++)  
    C[i] = A[i]*A[i] + B[i];
```

Assembly Code:

```
loop: ld  f1, 0(r1)  
      ld  f2, 0(r2)  
      fmul f1, f1, f1  
      fadd f1, f1, f2  
      st  f1, 0(r3)  
      addi r1, r1, 4  
      addi r2, r2, 4  
      addi r3, r3, 4  
      bne r3, r4, loop
```

Problem 3.A

Schedule operations into the VLIW instructions in the following table. Show only one iteration of the loop. Make the code efficient, but do not use software pipelining or loop unrolling. You do not need to write in NOPs (can leave blank).

ALU	Memory Unit	FPU

What performance did you achieve? (in FLOPS per cycle): _____

Problem 3.B

Unroll the loop by one iteration (so two iterations of the original loop are performed for every branch in the new assembly code). You only need to worry about the steady-state code in the core of the loop (no epilogue or prologue). Make the code efficient, but do not use software pipelining. You do not need to write in NOPs (can leave blank).

ALU	Memory Unit	FPU

What performance is achieved now? (in FLOPS per cycle): _____

Problem 2

Assume that the following code has been executed on a CPU with SCOREBOARD (no ROB).

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2	1	2	7	8
I2: ADDD F2 F6 F4	2	9	11	12
I3: MULTD F0 F4 F2	3	13	43	44
I4: SUBD F12 F2 F6	4	9	11	12
I5: ADDD F0 F12 F2	13	17	19	20

- A. List all the possible conflicts in the code.
- B. Is there a "configuration" that can respect the shown execution?
- If not, list at least 2 reasons for that and correct the table.
 - If it is possible, describe the architecture (e.g. how many units? Which kind? What latency?)

Answer 2.A

Answer 2.B

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2				
I2: ADDD F2 F6 F4				
I3: MULTD F0 F4 F2				
I4: SUBD F12 F2 F6				
I5: ADDD F0 F12 F2				

Problem 3

1. To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

State	Cache up to date?	Memory up to date?	Others have a copy?
Modified			
Exclusive			
Shared			
Invalid			

2. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 0	Exclusive (0)	Invalid	Yes	Yes
1	P1: write block 0				
2	P2: write block 0				
3	P2: write block 1				
4	P1: read block 1				
5	P2 : read block 1				
6	P1: write block 1				
7	P2: write block 1				
8	P1: write block 1				
9	P2: read block 0				

Question 1

- 1.** Explain the concept of simultaneous multithreading and describe advantages and disadvantages
- 2.** Describe the modification necessary to a multiple-issue processor to allow the management of simultaneous multithreading.

Question 2

1. Explain the difference between static scheduling and dynamic scheduling.
2. Explain the concept of trace scheduling and how it can be useful in VLIW machines

Question 3

In real processors, some architectural limits do not allow fully exploitation of the ILP present in the program. Specify these architectural limits and specify how they impact on the potential ILP.

